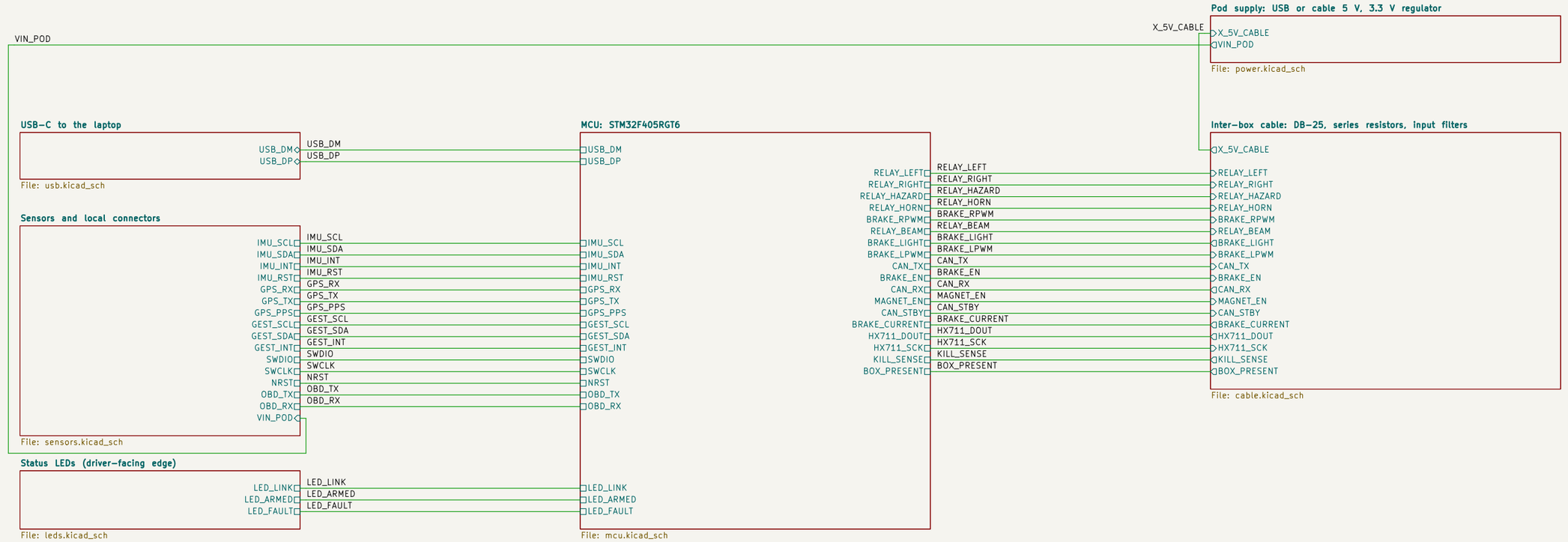
WINDSCREEN POD BOARD: block diagram. Each box is a sheet (double-click it in KiCad).
Power rails +3V3, VBUS and GND join the sheets by name (power symbols);
every other connection between sheets is a wire here.



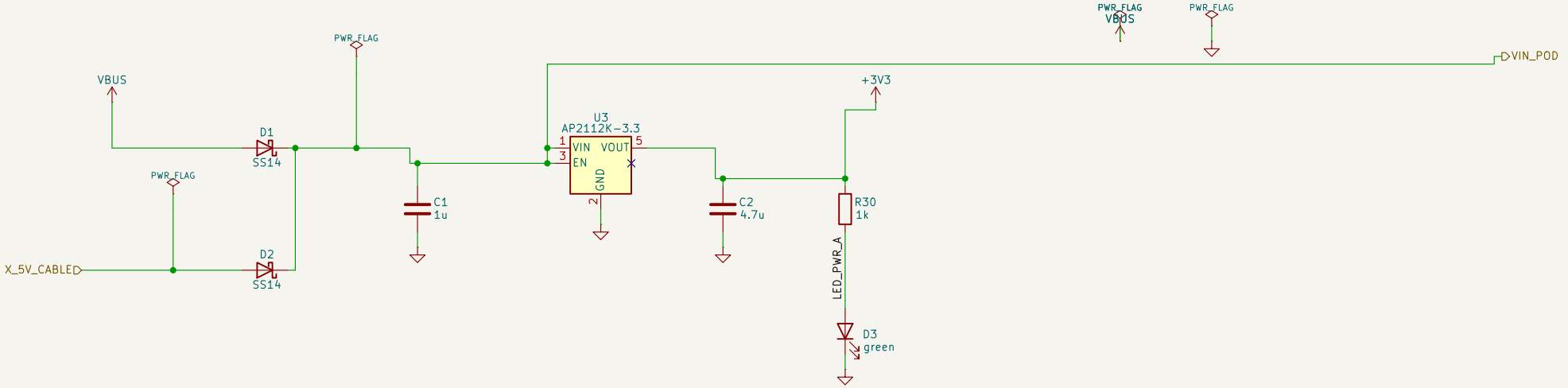
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /
File: pod_board.kicad_sch

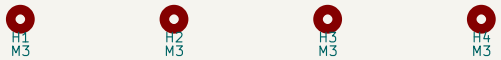
Title: Windscreen pod board (MCU, sensors, USB, cable)

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 1/7

POD SUPPLY: two 5 V sources, diode-OR'd: the laptop's USB (D1) and the power box through the cable (D2). Either one runs the pod, so it can be programmed and tested on USB alone. U3 makes 3.3 V for the MCU and sensors. D3: 3.3 V present.



Mounting holes: M3, no electrical connection



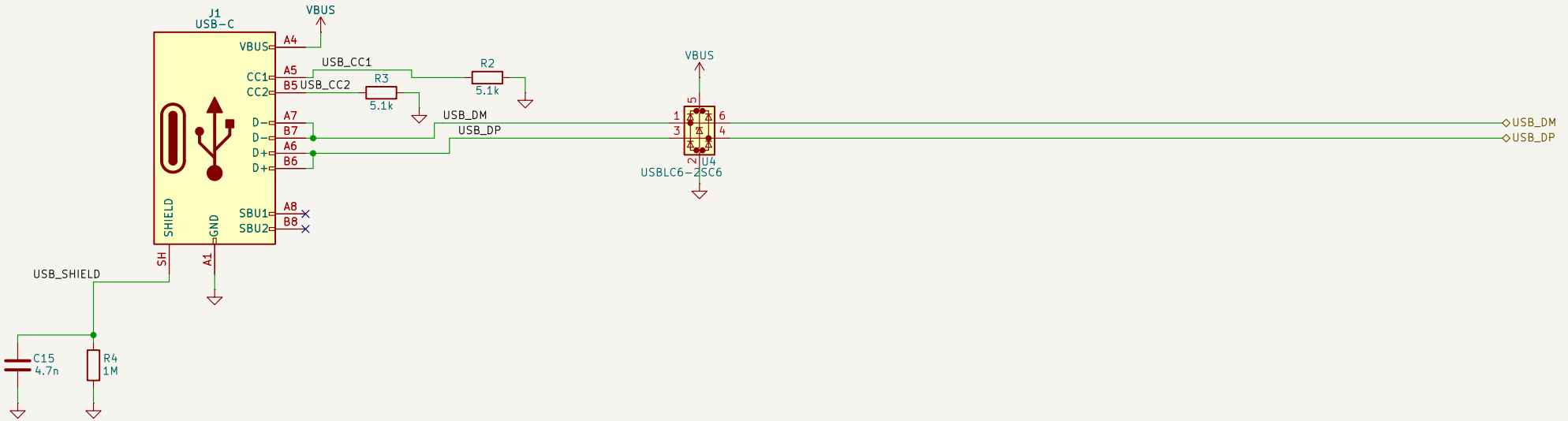
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Pod supply: USB or cable 5 V, 3.3 V regulator/
File: power.kicad_sch

Title: Pod supply: USB or cable 5 V, 3.3 V regulator

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 2/7

USB-C: the Part 3 link to the laptop, and the pod's power when the cable is out.
R2/R3 (5.1 k on CC) tell the laptop this is a device that wants 5 V. U4 clamps ESD on D+/D-
and VBUS. The shell goes to ground through R4/C15, so it drains static without a ground loop.



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /USB-C to the laptop/
File: usb.kicad_sch

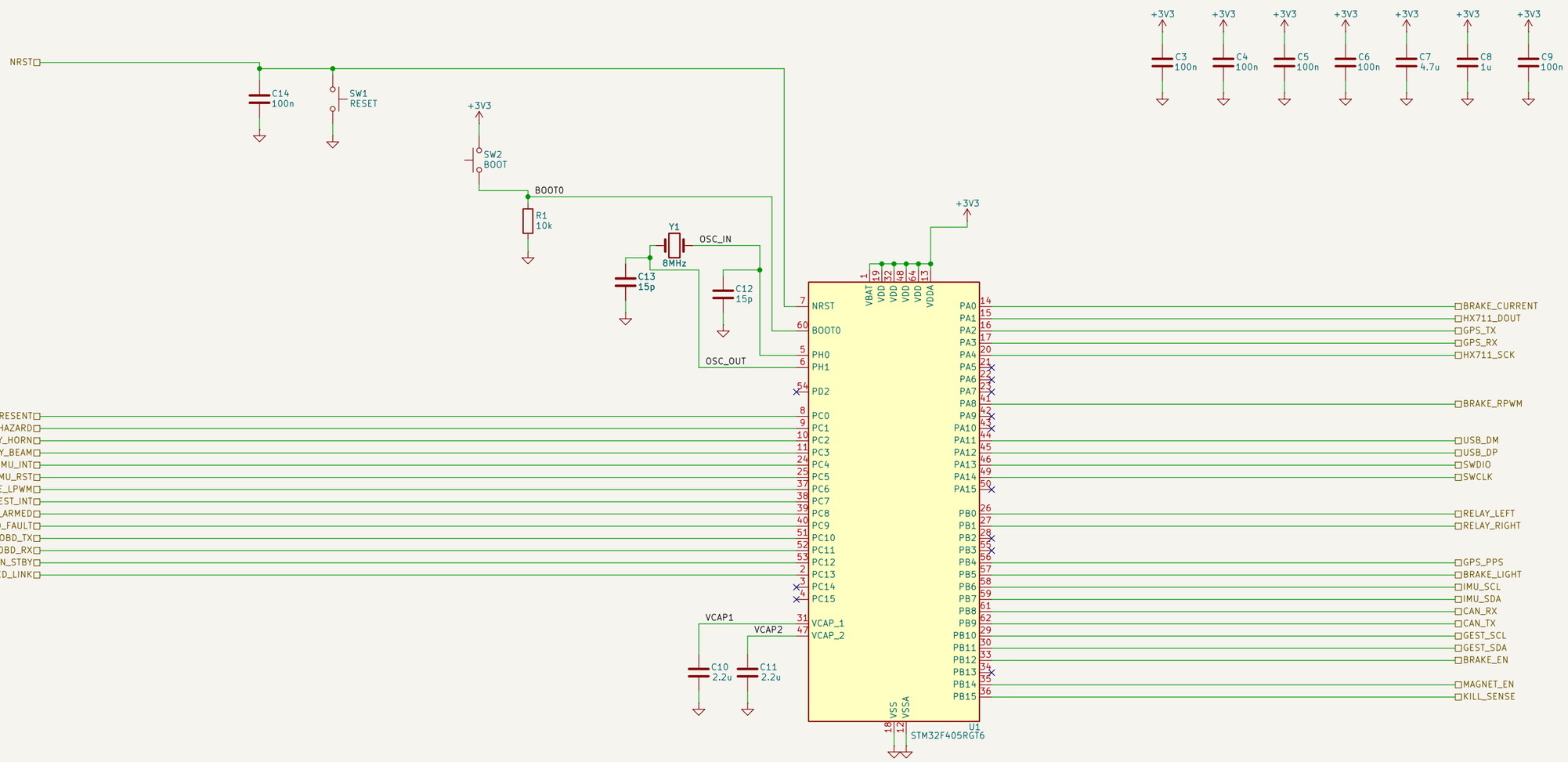
Title: USB-C to the laptop

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 3/7

MCU. Every pin assignment equals firmware Config.h (checked by check_nets.py). Each pin runs straight out to the sheet that uses it. Left: reset, boot, the 8 MHz crystal, the core regulator's capacitors (VCAP), and port C. Right: ports A and B.

SW1 RESET; SW2 BOOT: hold at reset for the USB DFU bootloader.
R1 holds BOOT0 low: run from flash.

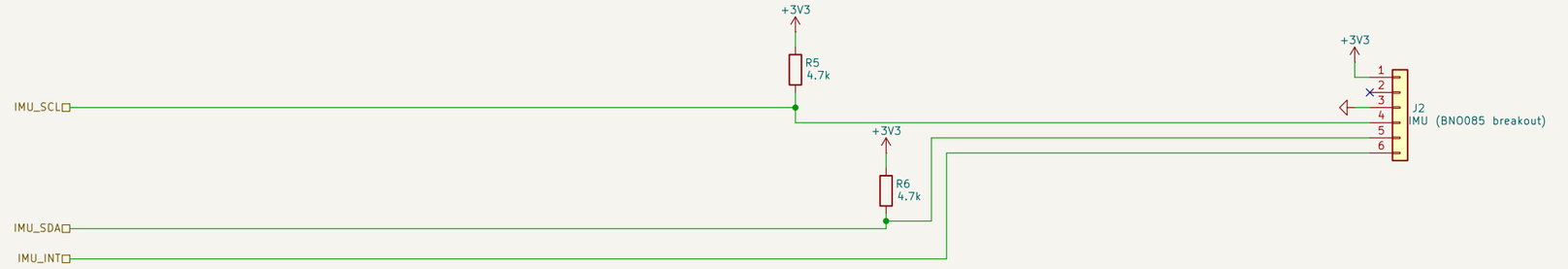
Decoupling: C3–C6 100 nF, one beside each VDD pin;
C7 4.7 uF bulk; C8 1 uF + C9 100 nF at VDDA.



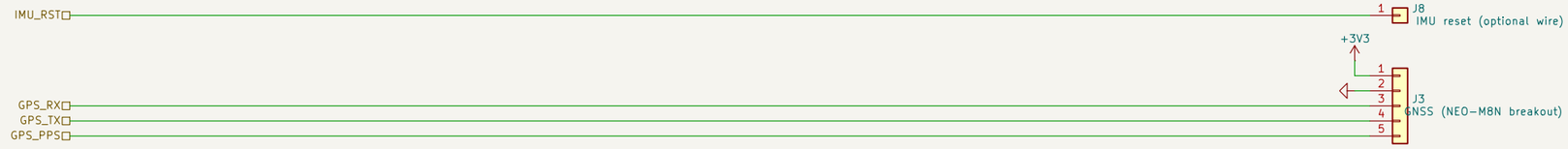
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /MCU: STM32F405RGT6/		
File: mcu.kicad_sch		
Title: MCU: STM32F405RGT6		
Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 4/7

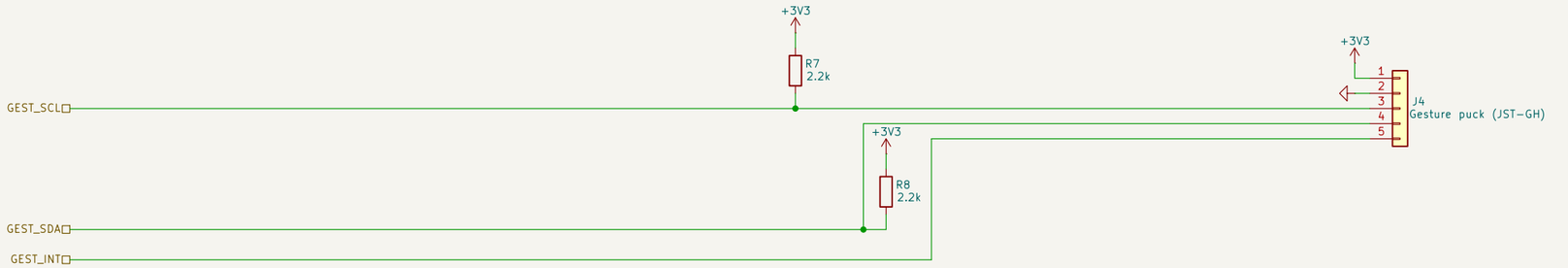
SENSORS AND LOCAL CONNECTORS. Breakouts on standoffs; MATCH EACH HEADER'S PIN ORDER TO THE BREAKOUT YOU BUY before layout. Names are the MCU's view (GPS_TX goes to the module's RX).



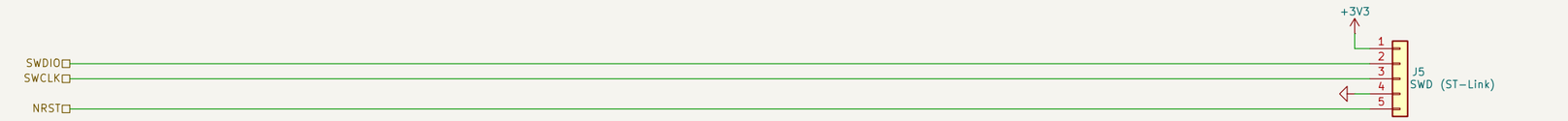
BNO085 IMU (I2C1, 4.7 k pull-ups R5/R6): J2 is pin for pin the Adafruit 4754's JP1 row (VIN 3Vo GND SCL SDA INT; 3Vo, its regulator output, left open).
J8: optional wire to the breakout's RST. Mount rigidly: it measures the car's motion.



NEO-M8N GNSS (USART2, PPS for timing).



Gesture puck: APDS-9960 on ~1.5 m of cable (I2C2 at 100 kHz; 2.2 k pull-ups for the cable).



SWD: ST-Link programming and debugging.



ELM327 fallback (USART3, HC-05 bridge):
build env hub_elm327 only (Part 4.8.5).
VCC is the pod's 5 V: the HC-05 board needs 3.6-6 V.

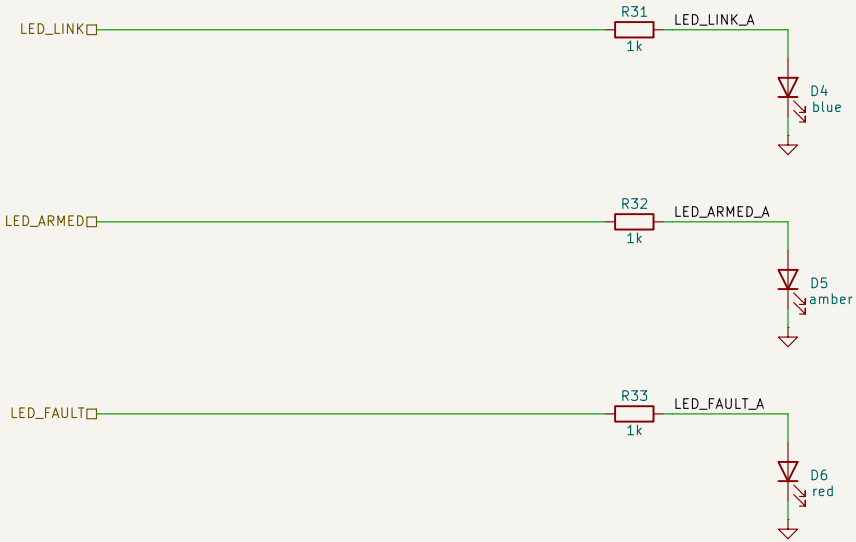
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Sensors and local connectors/
File: sensors.kicad_sch

Title: Sensors and local connectors

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 5/7

STATUS LEDs on the board's top edge, facing the driver: LINK (blue) the laptop is talking, ARMED (amber) the hub accepts brake commands, FAULT (red) a latched fault (Part 4.5).



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Status LEDs (driver-facing edge)/
File: leds.kicad_sch

Title: Status LEDs (driver-facing edge)

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 6/7

INTER-BOX CABLE (BUILD_GUIDE 4.8.3). MCU side on the left, the DB-25 on the right.
Outputs: 1 k in series (330 R on CAN_TX), so a pod on USB cannot push more than ~3 mA into an unpowered power box. Inputs: a 10 k pull-up on the cable side gives the SAFE reading with the cable out (kill engaged, box absent, brake light off, CAN recessive); 1 k + 10 nF filter PWM noise. BRAKE_CURRENT: 1 k + 100 nF (1.6 kHz) removes the 20 kHz ripple; 100 k reads 0 V with the cable out.

X_5V_CABLE

RELAY_LEFT — R44 1k — X_RELAY_LEFT
RELAY_RIGHT — R45 1k — X_RELAY_RIGHT
RELAY_HAZARD — R46 1k — X_RELAY_HAZARD
RELAY_HORN — R47 1k — X_RELAY_HORN
BRAKE_RPWM — R40 1k — X_BRAKE_RPWM
RELAY_BEAM — R48 1k — X_RELAY_BEAM

+3V3 — R59 10k — X_BRAKE_LIGHT
BRAKE_LIGHT — R58 1k — X_BRAKE_LIGHT
C60 10n

BRAKE_LPWM — R41 1k — X_BRAKE_LPWM
CAN_TX — R49 330 — X_CAN_TX
BRAKE_EN — R42 1k — X_BRAKE_EN

+3V3 — R62 10k — X_CAN_RX
CAN_RX — R61 100 — X_CAN_RX
MAGNET_EN — R43 1k — X_MAGNET_EN
CAN_STBY — R50 1k — X_CAN_STBY
BRAKE_CURRENT — R64 1k — X_BRAKE_CURRENT
C65 100n
R66 100k
HX711_DOUT — R63 100 — X_HX711_DOUT
HX711_SCK — R51 1k — X_HX711_SCK

+3V3 — R53 10k — X_KILL_SENSE
KILL_SENSE — R52 1k — X_KILL_SENSE
C54 10n

+3V3 — R56 10k — X_BOX_PRESENT
BOX_PRESENT — R55 1k — X_BOX_PRESENT
C57 10n

J7
DB-25 to power box
1 14 2 15 3 16 4 17 5 18 6 19 7 20 8 21 9 22 10 23 11 24 12 25 13
SH

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP — Kiogora Ian Mwenda
Sheet: /Inter-box cable: DB-25, series resistors, input filters/
File: cable.kicad_sch

Title: Inter-box cable: DB-25, series resistors, input filters

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 7/7